WORKED SOLUTION

STEP II 2014, Question 13

Probability and Statistics • Discrete Random Variables

STEP II 2014, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q13

The key idea

This is the birthday problem in disguise. The event $X \geq k$ just says that the first $k-1$ integers are all different, and $X = r$ says that the first $r-1$ are different and the rth matches one of them. Both probabilities are products of simple factors. The two curious identities then come from computing the same quantity in two ways: total probability equals 1 in part (i), and two formulas for $\mathrm{E}(X)$ agree in part (iv).

Solution

Throughout, write

$$Q_m = \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{m-1}{n}\right) \qquad (m \geq 1),$$

with $Q_1 = 1$ (an empty product). Since each integer is independent of the earlier ones and uniform on $\{1,\ldots,n\}$, Q_m is the probability that $I_1,\ldots,I_m$ are all different: given that the first j are distinct, the next avoids all of them with probability $1-\frac{j}{n}$.

The value $\mathrm{P}(X=4)$

We need I_1, I_2, I_3 distinct and then I_4 equal to one of those three:

$$\mathrm{P}(X=4) = 1\cdot\frac{n-1}{n}\cdot\frac{n-2}{n}\cdot\frac{3}{n} = \frac{3(n-1)(n-2)}{n^3}.$$

Part (i)

In the same way, $X = r$ means that $I_1,\ldots,I_{r-1}$ are distinct (probability Q_{r-1}) and I_r is one of these $r-1$ values (probability $\frac{r-1}{n}$, independently). So

$$\mathrm{P}(X=r) = \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{r-2}{n}\right)\frac{r-1}{n} = Q_{r-1}\,\frac{r-1}{n}, \qquad 2 \leq r \leq n+1.$$

Ask yourself Why do these probabilities add up to 1?

X certainly exists and lies between 2 and $n+1$: the first integer can't be a repeat, so $X \geq 2$; and among $I_1, \ldots, I_{n+1}$ there are $n+1$ integers taking only n values, so by the pigeonhole principle some value has appeared twice by time $n+1$, so $X \leq n+1$. (The chance that the sequence never repeats is zero, so nothing is missing.) Hence

$$\sum_{r=2}^{n+1} \mathrm{P}(X = r) = 1.$$

Writing out the terms $r = 2, 3, 4, \ldots$ gives

$$\frac{1}{n} + \left(1 - \frac{1}{n}\right)\frac{2}{n} + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\frac{3}{n} + \cdots = 1,$$

which is the required identity. The series stops at $r = n+1$, and any further terms are zero because they contain the factor $1 - \frac{n}{n}$. Done. With $m = r - 1$, this reads

$$\sum_{m=1}^{n} \frac{m}{n} Q_m = 1. \qquad (*)$$

Part (ii)

By definition of expectation,

$$\mathrm{E}(X) = \sum_{r=2}^{n+1} r \, \frac{r-1}{n}\left(1 - \frac{1}{n}\right)\cdots\left(1 - \frac{r-2}{n}\right) = \sum_{m=1}^{n} \frac{m(m+1)}{n} Q_m.$$

Part (iii)

$X \geq k$ happens exactly when there is no repeat among $I_1, \ldots, I_{k-1}$, that is, when these $k-1$ integers are all different. So

$$\mathrm{P}(X \geq k) = Q_{k-1} = \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{k-2}{n}\right), \qquad 2 \leq k \leq n+2,$$

and $\mathrm{P}(X \geq 1) = \mathrm{P}(X \geq 2) = 1$. (For $k = n+2$ the product contains $1 - \frac{n}{n} = 0$, as it should.)

Part (iv)

Write $p_r = \mathrm{P}(Y = r)$. Then $r = \sum_{k=1}^{r} 1$, so

$$\mathrm{E}(Y) = \sum_{r=1}^{N} r\, p_r = \sum_{r=1}^{N}\sum_{k=1}^{r} p_r = \sum_{k=1}^{N}\sum_{r=k}^{N} p_r = \sum_{k=1}^{N} \mathrm{P}(Y \geq k).$$

Swapping the order of summation is fine because the sum is finite: both double sums run over all pairs (k, r) with $1 \leq k \leq r \leq N$. Done.

Ask yourself Which two expressions for $\mathrm{E}(X)$ should we compare?

Apply this result to $Y = X$, which takes values in $\{1, \ldots, N\}$ with $N = n+1$ (the value 1 just has probability 0). Using part (iii),

$$\mathrm{E}(X) = \sum_{k=1}^{n+1} \mathrm{P}(X \geq k) = 1 + \sum_{k=2}^{n+1} Q_{k-1} = 1 + \sum_{m=1}^{n} Q_m.$$

Now set this equal to part (ii), splitting $\frac{m(m+1)}{n} = \frac{m^2}{n} + \frac{m}{n}$ and using $(*)$:

$$\sum_{m=1}^{n} \frac{m(m+1)}{n} Q_m = \sum_{m=1}^{n} \frac{m^2}{n} Q_m + \underbrace{\sum_{m=1}^{n} \frac{m}{n} Q_m}_{=1 \text{ by } (*)}.$$

So $1 + \sum_{m=1}^{n} Q_m = \sum_{m=1}^{n} \frac{m^2}{n} Q_m + 1$, that is,

$$\sum_{m=1}^{n} Q_m \left(1 - \frac{m^2}{n}\right) = 0.$$

The terms $m = 1, 2, 3$ are $\left(1 - \frac{1^2}{n}\right)$, $\left(1 - \frac{1}{n}\right)\left(1 - \frac{2^2}{n}\right)$ and $\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\left(1 - \frac{3^2}{n}\right)$, so this is exactly the required identity. As in part (i), every term beyond $m = n$ has a factor $1 - \frac{n}{n} = 0$. Done.

Checks

- For $n = 1$, $X = 2$ for certain. Part (i) gives $\frac{1}{1} = 1$, and the final sum is the single term $1 - 1 = 0$.
- For $n = 4$, the final sum is $\frac{3}{4} + \frac{3}{4} \cdot 0 + \frac{3}{4} \cdot \frac{1}{2} \cdot \left(-\frac{5}{4}\right) + \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{1}{4} \cdot (-3) = \frac{24-15-9}{32} = 0$, and both formulas give $\mathrm{E}(X) = \frac{103}{32}$.
- For $n = 365$ (birthdays), $\mathrm{E}(X) = 1 + \sum Q_m \approx 24.62$. A computer simulation of 10^5 sequences gave an average of 24.61, and exact enumeration for $n \leq 5$ confirmed every formula above.

Common slips

- Writing $\mathrm{P}(X = r)$ with $r - 1$ factors of the form $1 - \frac{j}{n}$ instead of $r - 2$. Only the second to the $(r-1)$th integers have to avoid earlier values.
- Forgetting the $k = 1$ term in $\sum \mathrm{P}(X \geq k)$. It contributes 1, which is exactly what cancels with $(*)$.

Beyond STEP

The tail-sum formula $\mathrm{E}(Y) = \sum_k \mathrm{P}(Y \geq k)$ holds for any random variable taking non-negative integer values, and its continuous version $\mathrm{E}(T) = \int_0^\infty \mathrm{P}(T > t)\,\mathrm{d}t$ is a standard tool in first-year probability. For large n, $\mathrm{E}(X) \approx \sqrt{\pi n/2}$, which is why collisions in hash functions turn up after only about $\sqrt{n}$ attempts (the "birthday attack"). Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.